MAZEN TAMER ELSAYED AHMED ELADAWY

Computer Science Undergraduate | Software Engineering Intern Candidate

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PROFILE

Computer Science undergraduate with a 3.55/4.00 GPA and hands-on work across algorithms, C# domain modeling, low-level programming, and game systems. Builds from first principles, documents technical decisions, and verifies results against source. Seeking a software engineering internship where careful problem solving and clear communication matter.

EDUCATION

October University for Modern Sciences and Arts (MSA), Egypt — Expected 2028
Bachelor of Science in Computer Science — **GPA 3.55 / 4.00**

TECHNICAL SKILLS

Languages: C++, C#, JavaScript, MIPS Assembly | **Platforms:** .NET 10, .NET Framework 4.7.2, Windows Forms, GDI+, HTML, CSS

Engineering: OOP, data structures, algorithms, binary I/O, bit packing, file persistence, generics, collections, events and delegates, collision detection, state machines | **Tools:** Git, GitHub, Visual Studio, MARS

SELECTED PROJECTS

Examination Management System — C# | .NET 10

- Modeled instructor and student workflows, three question types, exam state, scoring, and subject-based text persistence.
- Used abstract exam/question hierarchies, constrained generics, and exam-start events; the current project builds successfully.

Huffman Compression Algorithm — C++ | Binary I/O

- Built frequency generation, compression, and decompression stages for TXT, BMP, WAV, and AVI binary inputs.
- Constructed Huffman trees, generated prefix codes, and stored a bit-packed payload with the table required for reconstruction.

Library Management System — C# | .NET 10

- Created book, member, librarian, and manager models with ID/title search, borrowing, and availability tracking.
- Kept borrowing as a member-owned state transition while the manager coordinates collection operations; the project builds successfully.

Windows Forms Platformer — C# | GDI+

- Implemented a custom timer-driven update loop, off-screen rendering, camera scrolling, collision, combat, health, and boss systems without a game engine.

Monoalphabetic Cipher — MIPS Assembly

- Implemented letter validation, duplicate-free key expansion, and reversible case-preserving substitution in MARS.

EXPERIENCE

IGCSE O Level Mathematics Tutor — 2023 - 2024

- Taught O Level mathematics through structured, step-by-step explanations adapted to individual student needs.

RELEVANT COURSEWORK

Data Structures, Algorithms, Operating Systems, Computer Architecture, Object-Oriented Programming, Web Development, Discrete Mathematics, Linear Algebra, Calculus I and II